PROJECT STATUS REPORT

KVL Live Translator

Real-time meeting assistant — English ↔ Hindi/Hinglish, built phase by phase

21 August 2026

11 of 15 phases complete (73%)

✓ COMPLETED — PHASES 1–11

1 **Architecture & UI Shell** DONE

pnpm monorepo (Electron + React + TS desktop, Node + WebSocket backend), navigable sidebar/screen shell, Platinum Silver & Gold theme established.

2 **Audio Capture Engine** DONE

Modular mic + system-audio capture, VAD-based chunking, device selection, real-time level meters, diagnostics screen.

3 **Speech-to-Text** DONE

Provider-based STT (Local Whisper → Cloud), encrypted key relay via Electron safeStorage, live transcript streaming.

4 **Translation** DONE

Local → Free → Paid translation router with confidence-based fallback, Hindi/Hinglish/Simple modes, glossary term protection.

5 **Live Translator UI** DONE

Full live meeting screen — dual transcript/translation panels, font size, compact mode, floating mini window.

6 **My Response Engine** DONE

Speak or type Hindi/Hinglish → reviewable professional English draft, 7 response styles, push-to-talk recording.

7 **Text-to-Speech Engine** DONE

Local (espeak-ng) → Browser → Cloud voice tiers, output device selection, virtual-cable routing guidance.

8 **Meeting Understanding Assistant** DONE

"What does this mean?" per-sentence explanations — simple meaning, what's being asked, key terms, suggested response, with selectable context windows.

9 Smart Meeting Copilot DONE

Automatic real-time detection of Questions, Decisions, Action Items, Deadlines, Requirements, Objections and more — free local detection plus batched cloud analysis, confirm/edit/delete workflow.

10 AI Meeting Summary DONE

11-section post-meeting summary with confirmed facts structurally separated from AI interpretation, Meeting History screen, TXT/Markdown/PDF/JSON export, cascade delete.

11 Multi-Provider Orchestration & Cost Optimization DONE

Usage/cost tracking, monthly budget enforcement, 3 cost modes (Economy/Balanced/Quality), Local-Only privacy mode, provider health monitoring, request caching, API Usage dashboard, Debug screen, provider Test-key.

○ REMAINING — PHASES 12–15

PHASE	NAME	WHAT IT COVERS
12	Meeting App Window Optimization	Always-on-top compact mode tuned for Zoom/Meet/Teams overlay use — building on the existing floating mini window from Phase 5.
13	Privacy & Security Hardening	Deeper audit and hardening pass across the app's privacy/security surface.
14	Production Polish	Long-meeting stability and crash recovery — partly already addressed by Phase 10's transcript-buffer fix and Phase 11's cache memory cap.
15	Final Release	Installer packaging and first-run setup wizard.

> **Note:** No detailed spec has been provided yet for Phase 12 onward. Every phase so far has been built, tested, and verified one at a time before moving to the next — that discipline continues once the next phase's requirements are shared.